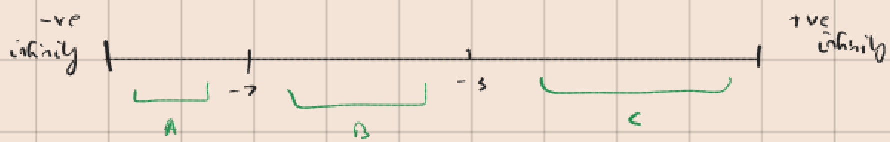
3) Rational Inequalities

$\dfrac{f(x)}{d(x)}$ $\square$ (sign of inequality) , $\dfrac{h(x)}{g(x)}$

Ratio of Polynomial

$$\frac{x+7}{x+3} > 0 \longrightarrow \frac{x+7}{x+3} = 0 \longrightarrow x+7=0 \quad , \quad x+3=0$$

$x = -7 \qquad x = -3$



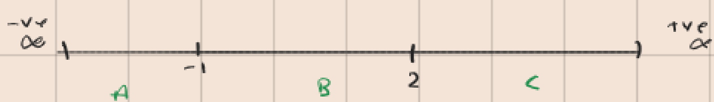
		test points	
A =	$x > -7$	-8	+
B =	$-7 < x < -3$	-4	—
C =	$x < -3$	-2	+

$x \in (-\infty, 7) \cup (-3, \infty)$

solution set

Q) $\dfrac{x+1}{x-2} \geq 0$

$x = -1 \qquad x = 2 \qquad$ (critical points)



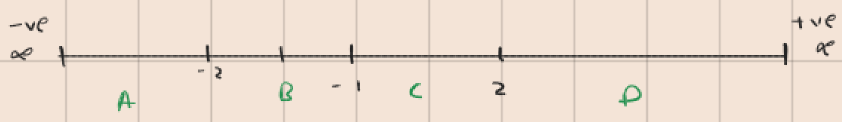
		test points	
A	$x < -1$	-2	+
B	$-1 < x < -2$	0	−
C	$x > 2$	4	+

$x \in (-\infty, -1] \cup (-2, \infty)$

Q) $\dfrac{x^2-4}{x+1} > 0$

$$\frac{(x+2)(x-2)}{x+1} > 0$$

$x = -2$, $x = 2$ $x = -1$ (critical points)



A	$x < -2$	-3	$-$
B	$-2 < x < -1$	1.5	$+$
C	$-1 < x < 2$	1	$-$
D	$x > 2$	3	$+$

$x \in (-2, -1) \cup (2, \infty)$

Absolute Inequality

$|x+1|$, $|x|$, $|x+5|$

Q) $\left|\frac{2x+2}{4}\right| + 4 \leq 5$

$\left|\frac{2x+2}{4}\right| \leq 1$

$\frac{2x+2}{4} \geq 1$ or $\frac{2x+2}{4} \leq 1$

$2x+2 \geq 4$ $\quad$ $2x+2 \leq 4$

$x+1 \geq 2$ $\quad$ $x+1 \leq 2$

$x \geq 1$ $\quad$ $x \leq 1$

Properties

$|x| < a$

$-a < x < a$ $\quad$ $(-a, a)$

$|x| \leq a$

$-a \leq x \leq a$ $\quad$ $[-a, a]$

$|x| > a$

$x > a$ or $x < -a$

$(a, \infty) \cup (-\infty, -a)$

$|x| \geq a$

$x \geq a$ or $x \leq a$

$[a, \infty) \cup (-\infty, a]$